

HRIS · WORKFORCE ANALYTICS

AI Employee Retention Intelligence System

Predicting flight-risk employees from workforce data — and turning that signal into a targeted retention strategy.

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Machine Learning Capstone Project

Turnover is expensive — and mostly invisible until it's too late

Challenges : HR teams typically react to attrition after it happens — exit interviews, replacement hiring — rather than identifying risk early.

Business question: Can we predict which employees are at risk of leaving, and explain why — early enough to intervene?

Objective: Build a classification model that flags flight-risk employees and surfaces the drivers behind their risk, so HR can act before resignation letters land.

16.1% of employees in this workforce leave — a costly minority to miss

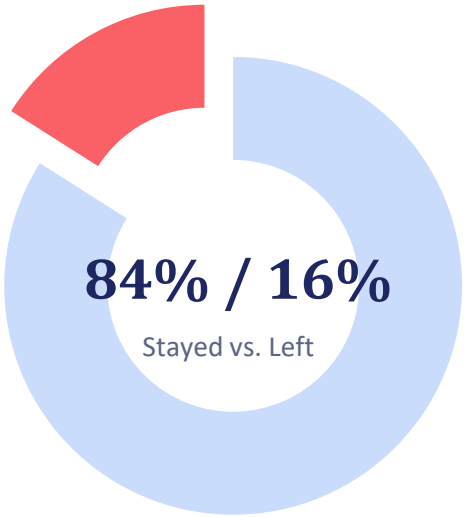
Why it matters

Every voluntary exit carries **recruitment cost, lost institutional knowledge, and operational disruption** — an early warning system turns reactive HR into proactive retention.

A clean, well-understood dataset — with one major challenge

1,470 Employee records	31 Predictive features, post-cleaning
0 Missing values or duplicates	4 Constant / non-informative columns dropped

The core challenge: severe class imbalance



Why it matters

A model that always predicts “stays” looks 84% accurate — while catching zero at-risk employees. Accuracy alone is a deceptive metric here; Recall, Precision and F1 drive every decision from here on.

Source: IBM HR Analytics Employee Attrition dataset

Five clear behavioral signals separate stayers from leavers



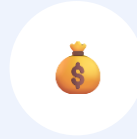
30% vs 10%

Attrition rate for
employees on mandatory
overtime vs. those without



Sales

highest-attrition
department, well above
R&D and HR



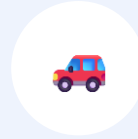
\$3.2k Vs 5.2k

median monthly income:
leavers vs. stayers



0-3 yrs

tenure window where
voluntary turnover is
heavily concentrated

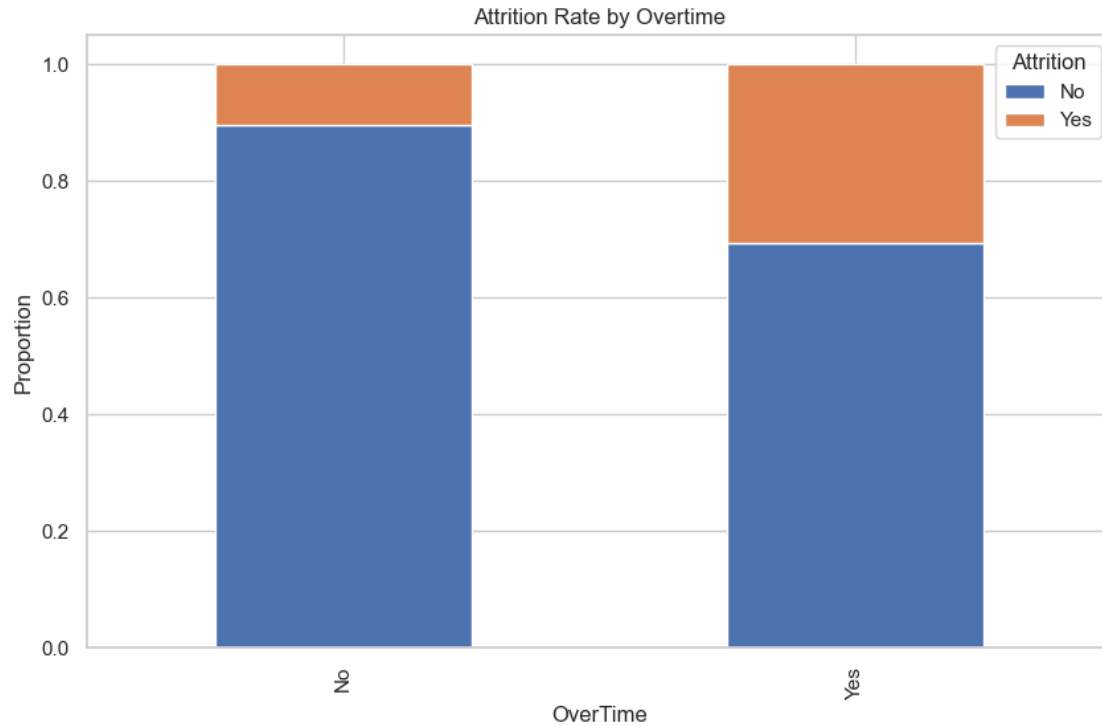


9 mi vs 7 mi

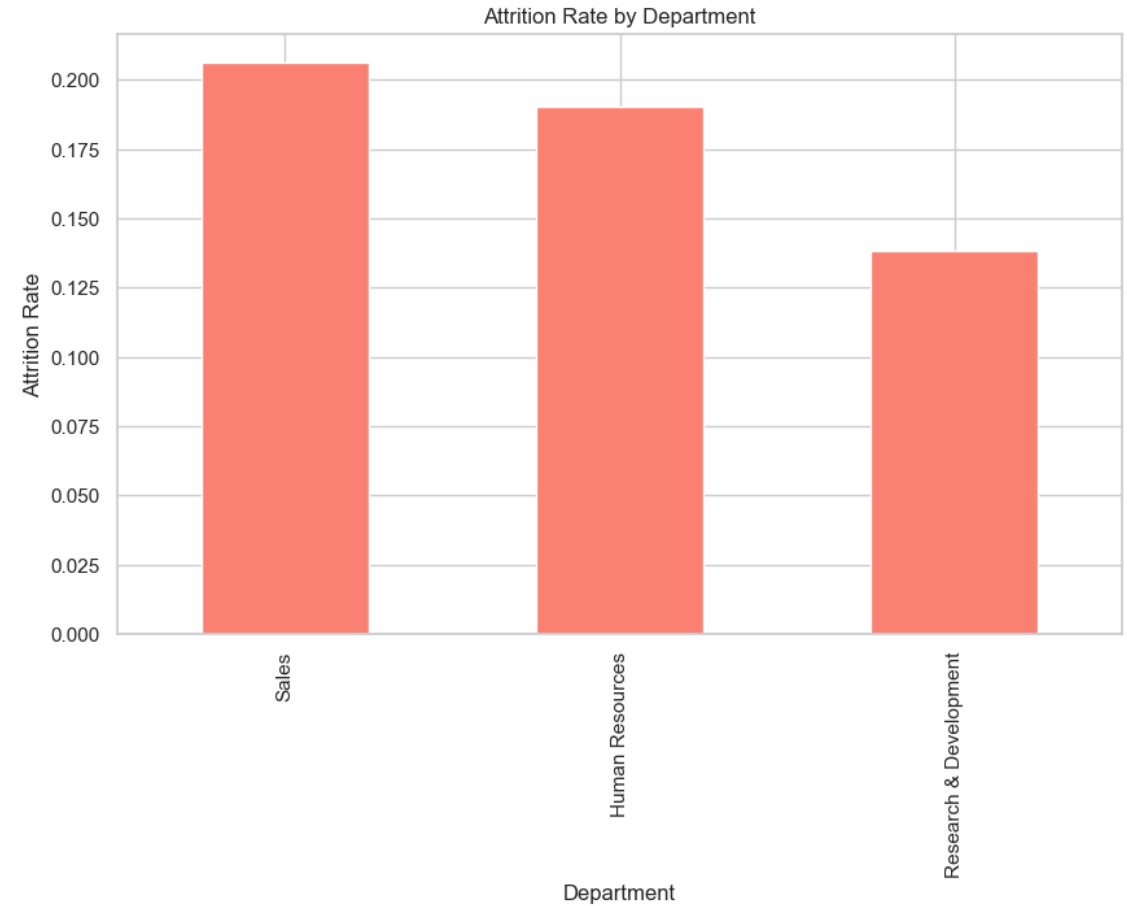
median commute distance:
leavers vs. stayers

Key Findings & Insights (1/2)

What drives attrition risk — the HR-analyst read



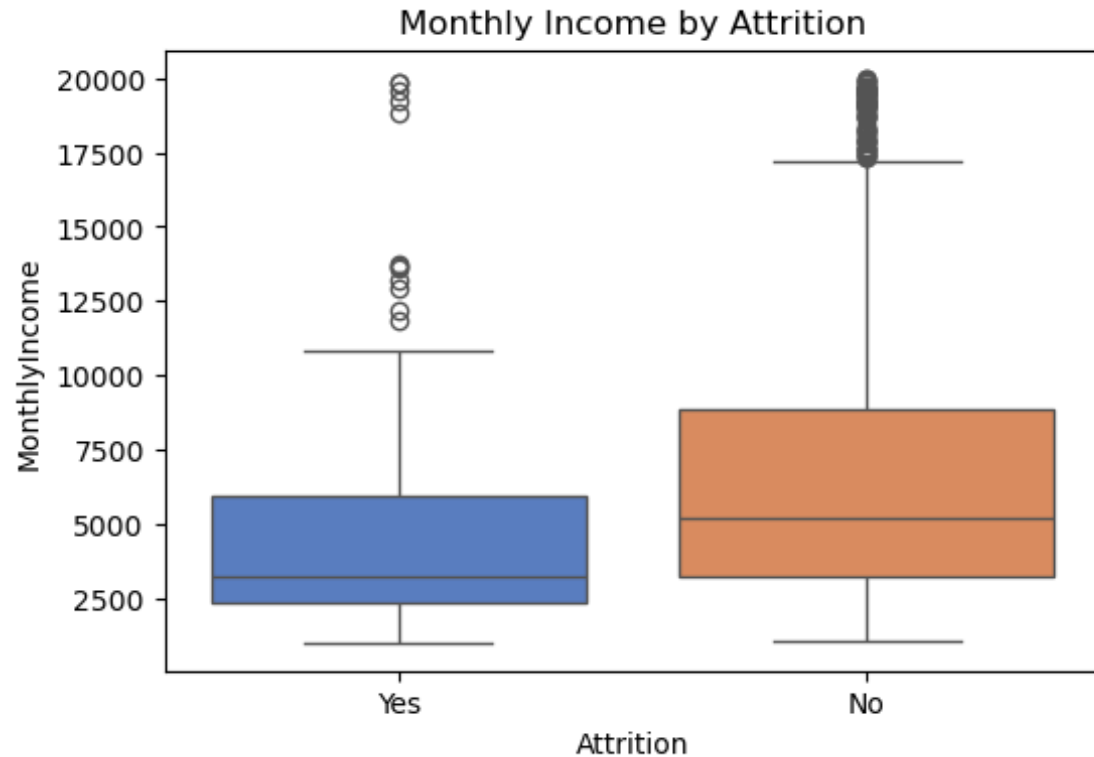
Overtime employees show a visibly higher attrition rate — a real, actionable burnout signal, not just noise.



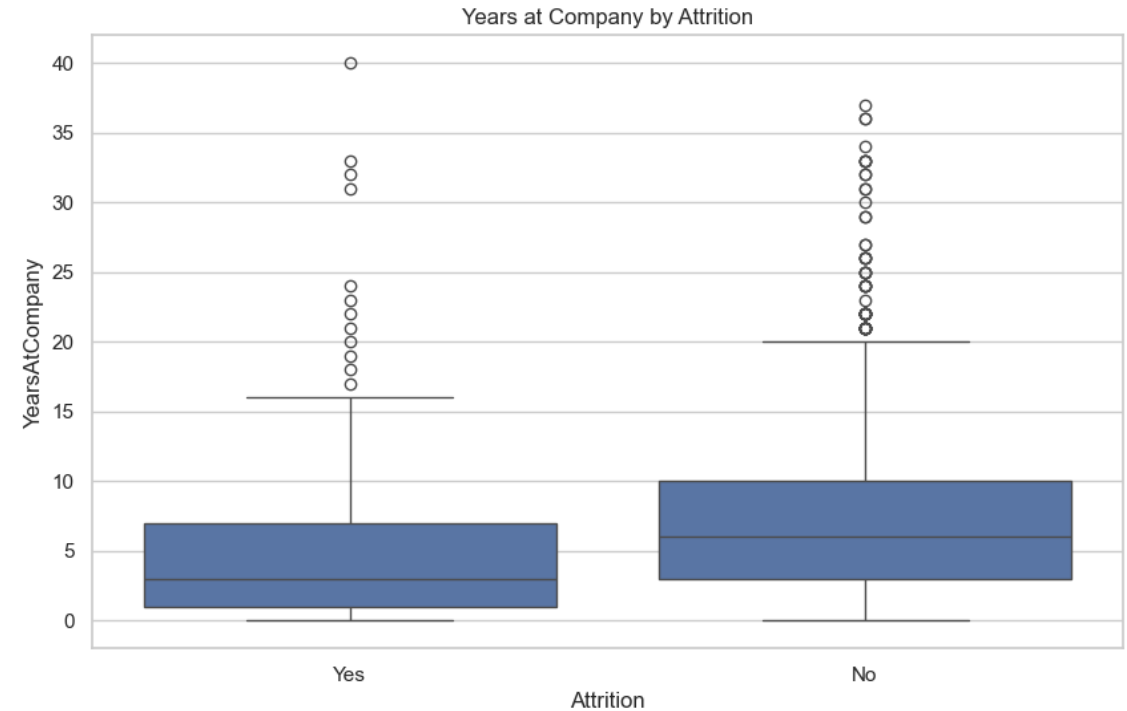
Attrition is not evenly spread across departments — one shows a meaningfully higher rate, pointing to team-level rather than company-wide risk.

Key Findings & Insights (2/2)

Compensation and tenure patterns



Employees who leave skew toward lower Monthly Income — supporting the Income Percentile feature over raw salary alone.



Attrition concentrates in shorter-tenure employees — the highest-risk window is early in the employee lifecycle, before full integration.

Turning workforce behavior into predictive, HR-specific signals

Burnout Index

`Overtime pts + Travel pts + (3 - Work-Life Balance)`

Combines scheduling strain, travel load, and self-reported balance into one composite risk score.

Career Stagnation Index

`(Years in Role + 1) ÷ (Years at Company + 1)`

Flags employees whose role hasn't evolved relative to their tenure — a proxy for growth stalling.

Preprocessing pipeline (leak-proof: fit on train, transform on test only)

1

Split

80/20 stratified



2

Encode

OneHotEncoder categoricals



3

Scale

StandardScaler numerics



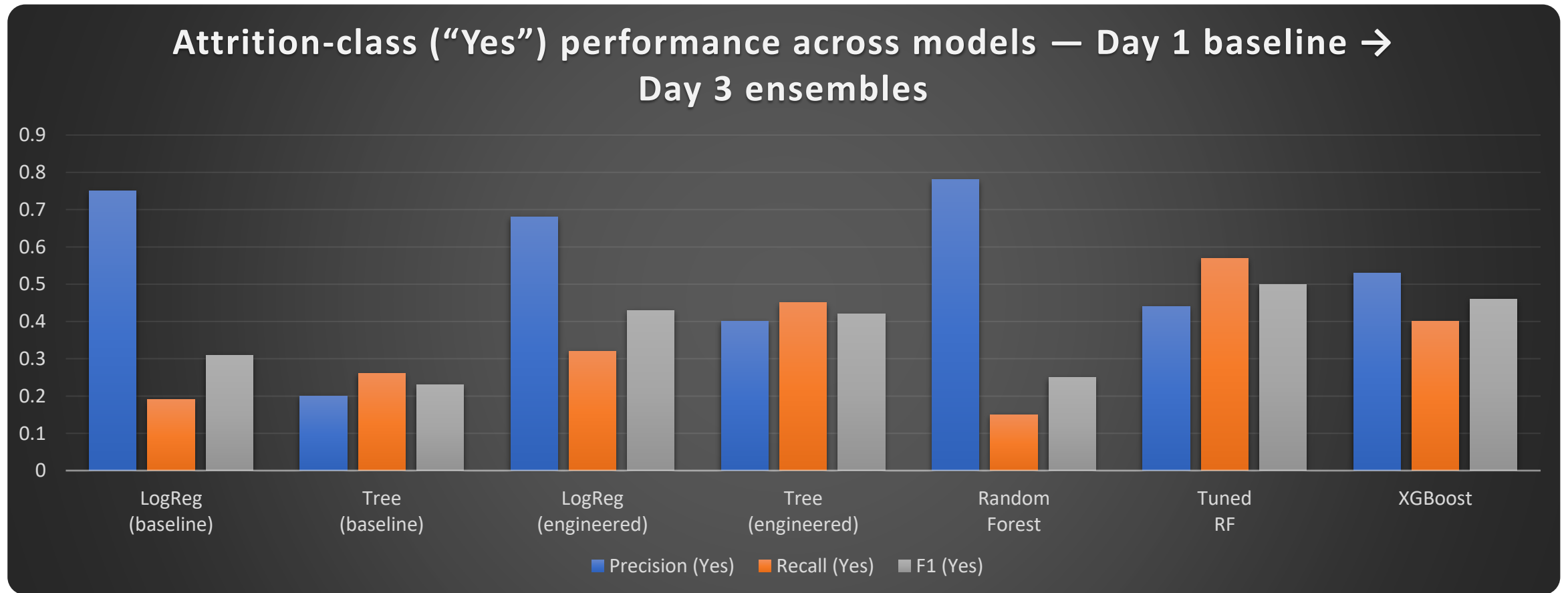
4

De-collinearize

Drop JobLevel,
Years With Current Manager,
Years In Current Role

Model Building & Evaluation

From baseline to full ensemble comparison — optimizing for Recall, not Accuracy



Key read: Tuned Random Forest is the strongest single ensemble model (Recall 0.57, F1 0.50) — but raw ensembles without tuning (Bagging, plain Random Forest, AdaBoost) barely beat the baseline on Recall despite higher accuracy.

A leak-proof pipeline crowned a stable, production-ready champion



GridSearchCV

swept `n_estimators`, `max_depth`, and `class_weight` across Random Forest configurations



Resampling trio

compared oversampling, undersampling, and SMOTE to correct the 84/16 imbalance



Stratified 5-Fold CV

validated the champion inside an imblearn Pipeline, so resampling never touched validation folds

CHAMPION MODEL

Oversampled Logistic Regression

72.11%

Mean cross-validated Recall
(± 0.07 across 5 folds)

Low fold-to-fold volatility mathematically confirms this ~70%+ catch rate is a structural property of the model — not a lucky split.

Choosing a Strategy

There is no single “best” model — only the best fit for the strategy

HIGH-COVERAGE STRATEGY

Tuned Random Forest

57% Recall / 44% Precision

Casts the widest net — catches more than half of all employees who leave. Best when broad, low-cost interventions (check-ins, surveys) are viable.

HIGH-CONFIDENCE STRATEGY

Tuned AdaBoost

72% Precision / 28% Recall

Flags fewer employees but is right more often. Best when interventions are expensive — e.g. retention bonuses — and false alarms are costly.

The takeaway: model choice is a business decision, not just a technical one — it should follow the cost of a missed leaver vs. the cost of a wasted intervention.

Hyperparameter Tuning & Model Optimization

GridSearchCV (scored on Recall) + imbalance-handling strategies

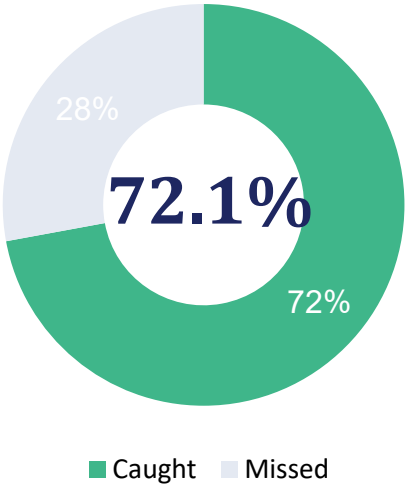
Model Configuration	Precision	Recall	F1
Oversampled Logistic Regression	0.407	0.702	0.516
SMOTE Logistic Regression	0.397	0.660	0.496
Undersampled Logistic Regression	0.364	0.681	0.474
Grid-Tuned Random Forest	0.389	0.596	0.471

✓ **CHAMPION MODEL: Oversampled Logistic Regression**

Wins on all 3 metrics simultaneously — a simpler technique outperformed SMOTE and a tuned Random Forest. Validated with leak-proof 5-fold cross-validation (resampling isolated inside each fold):

Recall 0.721 ± 0.072 | **Precision** 0.380 ± 0.034 | **F1** 0.497 ± 0.037

Champion Model:
At-Risk Employees Caught



Cross-validated on 5 folds using an imblearn Pipeline — oversampling recomputed fresh inside each fold to prevent leakage.

From model output to HR action



Risk scoring at scale

Flag employees for proactive 1:1 check-ins before resignation, prioritized by predicted risk.



Targeted retention spend

Direct limited retention-bonus budget toward the high-precision segment, not company-wide raises.



Root-cause guidance

Burnout Index and overtime signal point HR toward workload and scheduling fixes, not just pay.

Ethical considerations & limits

A risk score must inform supportive conversations, never punitive ones — flagging someone as “flight risk” can itself damage trust if mishandled. The model reflects one company's data (1,470 records) and may not generalize across industries, tenure structures, or regions. It should complement manager judgment, not replace it, and be revisited regularly as workforce dynamics shift.

Challenges → Solutions

Three moments that shaped the final model — problem, discovery, fix

Severe Class Imbalance

1

PROBLEM

84% / 16% split — accuracy was meaningless

FIX

Reframed around Recall/Precision/F1 from Day 1; compared oversampling, undersampling, and SMOTE side-by-side

AdaBoost Nearly Collapsed

2

PROBLEM

Default settings → 2% recall on the minority class

FIX

Deeper base trees + higher learning rate recovered performance to 28% recall

Cross-Validation Caught a Leak

3

PROBLEM

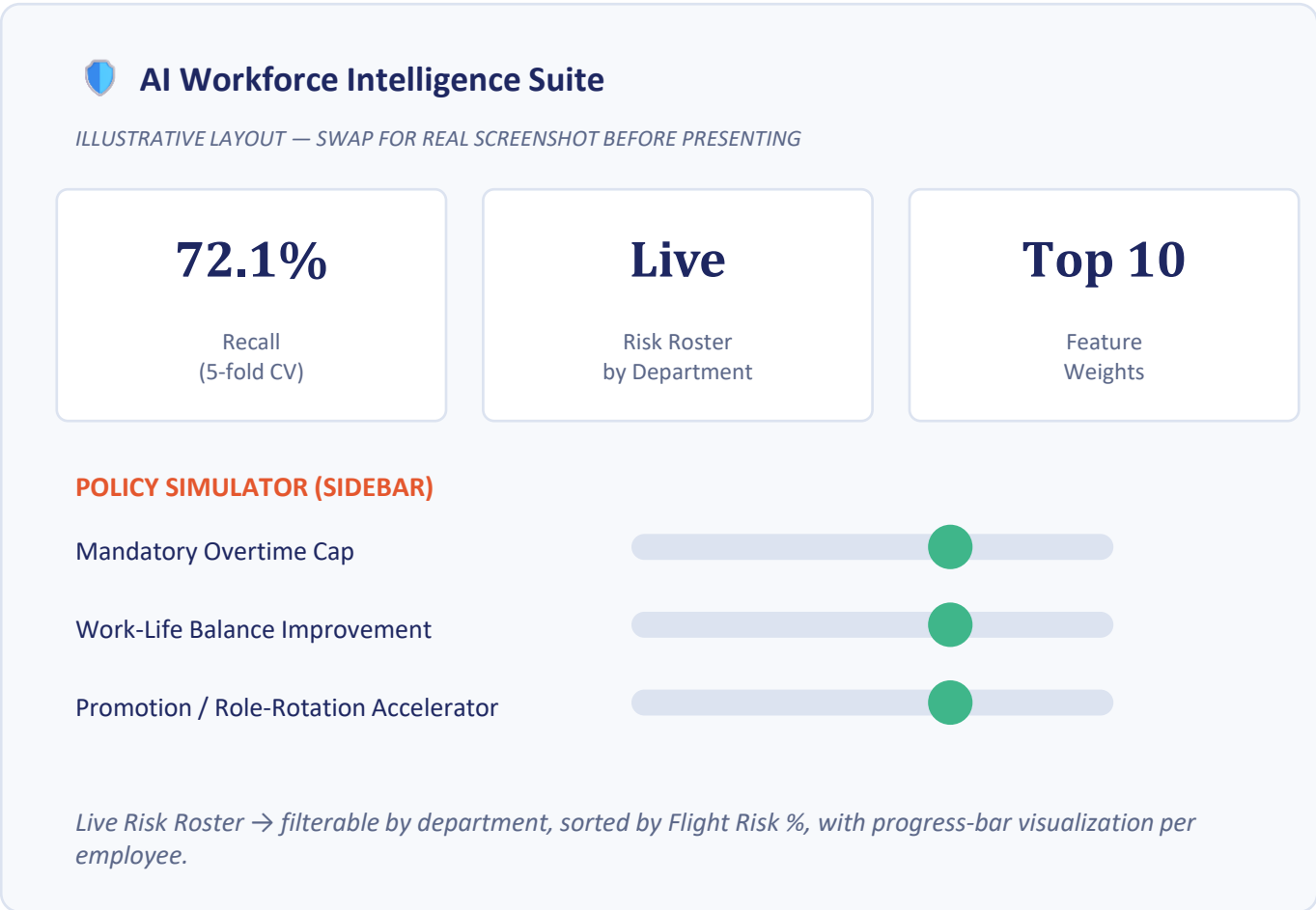
First CV run inverted precision & recall vs. the single test split

FIX

Rebuilt with an imblearn Pipeline — leak-proof, confirmed 72% recall

Live Demo — AI Workforce Intelligence Suite

Extended into an interactive Streamlit tool — already built, not just planned



⚠ HONESTY NOTE

The app's “Net Corporate Savings” figure uses an illustrative assumption (35% intervention success × \$25k/replacement) — not derived from this dataset. Labeled explicitly in-app, not presented as fact.

WHAT'S NEXT

- SHAP explainability per employee
- Validate across multiple years / companies
- Pilot with a live HR team ➡ [Live Simulator](#)

Thank you

AI Employee Retention Intelligence System

Questions & discussion welcome.

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